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Question Paper Code: 11060

B.E/B.Tech. DEGREE EXAMINATIONS, APRIL / MAY 2025

Fourth Semester

Computer Science and Engineering

(Common to Information Technology)

22CS492 – Database Management Systems

(Regulation 2022)

Time: Three hours

Maximum:100 marks

Answer ALL questions.

PART - A ($10 \times 2 = 20$ marks)

1. What is a data model?
2. Describe the function of schemas in a database system.
3. What are constraints in the relational model?
4. Outline the purpose of SQL UPDATE and the conditions under which it can be used.
5. Define a database view.
6. Differentiate between views and tables in a database.
7. List the different types of schedules in transaction processing.
8. What are the advantages and disadvantages of two-phase locking?
9. What is data replication in distributed databases?
10. Describe how NoSQL databases achieve high availability.

PART - B ($5 \times 13 = 65$ marks)

11. a. Design an Enhanced - ER model for an online shopping platform that includes the following entities:
 - Customer with attributes customer_id, name, email, address.
 - Product with attributes product_id, name, price.
 - Order with attributes order_id, order_date, order_status.
 - Implement complex relationships such as specialization (e.g., VIP and Regular customers) and generalization (e.g., Electronics and Clothing categories for products).

(13)

(OR)

11. b. Explain the process of data modeling using the ER model. Provide examples of how entities, relationships, and attributes are represented.

(13)

12. a. Write SQL queries to:
- (i) Create a table Employee with columns emp_id, emp_name, emp_salary, and emp_department_id (including necessary constraints).
 - (ii) Insert at least three records into the Employee table.
 - (iii) Update the salary of an employee based on their emp_id.
 - (iv) Delete an employee record based on a given condition. (13)
- (OR)
12. b. Explain the difference between centralized and client-server DBMS architectures. Using a university's student information system, propose an architecture for both models and explain how each model would handle: Data storage and access, User queries, Data security and backup, Scalability and performance. (13)
13. a. Explain the use of JDBC in database connectivity. Provide a step-by-step process for connecting a Java application to a database using JDBC. (13)
- (OR)
13. b. Describe the challenges in database programming. How do techniques like stored procedures, triggers, and views help in overcoming these challenges? (13)
14. a. Analyze the role of ACID properties in ensuring reliable database transactions. Provide real-world examples to illustrate their importance. (13)
- (OR)
14. b. Discuss the need for database recovery mechanisms. Analyze how different recovery techniques (deferred update, immediate update) ensure data consistency. (13)
15. a. Compare different data fragmentation techniques and evaluate their impact on database performance. (13)
- (OR)
15. b. Examine the role of NoSQL databases in handling unstructured data. Analyze their impact on modern database architectures. (13)

PART - C (1×15 = 15 marks)

16. a. Evaluate the importance of functional dependencies in database normalization. How do they impact redundancy and consistency? (15)
- (OR)
16. b. Evaluate the role of Boyce-Codd Normal Form (BCNF) in database design. How does it differ from 3NF, and when is it necessary? (15)

Question Paper Code: 11090

B.E/B.Tech. DEGREE EXAMINATIONS, APRIL / MAY 2025

Fourth Semester

Computer Science and Engineering

22CS494 – Computer Networks

(Regulation 2022)

Time: Three hours

Maximum:100 marks

Answer ALL questions.

PART - A ($10 \times 2 = 20$ marks)

1. What are the three criteria necessary for an effective and efficient network?
2. Compare OSI and TCP.
3. Recall why is flow control and error control duplicated in different layers?
4. Contrast between stop and wait and sliding window protocol.
5. Define subnetting.
6. Infer what is the network address in a class A subnet with the IP addresses of one of the hosts as 25.34.12.56 and mask 255.255.0.0?
7. How IANA has divided port numbers?
8. Show the four aspects related to the reliable delivery of data?
9. Why do we need a Domain Name System? What role does the DNS Resolver play in the DNS system?
10. Outline the Headers and methods of HTTP.

PART - B ($5 \times 13 = 65$ marks)

11. a. With the functionalities of every layer draw and explain the OSI network architecture in details. Give some advantages and disadvantages of combining the session, presentation, and application layer in the OSI model into one single application layer in the Internet model. (13)
(OR)
11. b. Make use of a light signal is travelling through a fiber. What is the delay in the signal if the length of the fiber-optic cable is 10 m, 100 m, and 1 Km (Assume a propagation speed of 2×10^8 m/s)? Also name the two major categories of transmission media. (13)
12. a. Assume, given a remainder of 111, a data unit of 10110011 and a divisor of 1001, is there an error in the data unit. Justify your answer with necessary principles. (13)
(OR)

12. b. Discover how hidden node and exposed node problem is solved in IEEE 802.11. (13)

13. a. Choose an organization which is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets.

A. Find the subnet mask. (3)

B. Find the number of addresses in each subnet. (3)

C. Find the first and last addresses in subnet 1. (3)

D. Find the first and last addresses in subnet 1024. (4)

(OR)

13. b. Make use of the following protocols in internet and explain in detail :

i) ICMP (5)

ii) ARP (4)

iii) RARP (4)

14. a. With neat architecture, explain TCP and its sliding window algorithm for flow control. (13)

(OR)

14. b. Identify whether the user send data at 1.2 Mbps all the time? Is it guaranteed that frames are never discarded in this case? If the answer is no, is it guaranteed that frames are discarded only if there is congestion? (13)

15. a. Analyze and explain about HTTP. Give their uses, state strengths and weaknesses. (13)

(OR)

15. b. Examine how the Simple Mail Transfer Protocol (SMTP) is useful in electronic mail. (13)

PART - C (1×15 = 15 marks)

16. a. Choose an IPv4 datagram, the *M* bit is 0, the value of HLEN is 5, the value of total length is 200, and the offset value is 200. What is the number of the first byte and number of the last byte in this datagram? Is this the last fragment, the first fragment, or a middle fragment? (15)

(OR)

16. b. A router on an Ethernet network has received a multicast IP packet with group id 226.17.18.4. When the host checks its multicast group table, it finds this address. Show how the router sends this packet to the recipients by encapsulating the IP packet in an Ethernet frame. Show all the entries of the Ethernet frame. The outgoing IP address of the router is 185.23.5.6, and its outgoing physical address is 4A224512E1E2. Does the router need the services of ARP? (15)

Question Paper Code: 11023

B.E/B.Tech. DEGREE EXAMINATIONS, APRIL / MAY 2025

Fourth Semester

Computer Science and Engineering

(Common to Information Technology)

22CS491 – Operating Systems

(Regulation 2022)

Time: Three hours

Maximum:100 marks

Answer ALL questions.

PART - A (10×2 = 20 marks)

1. State the advantages of multiprocessors.
2. Define Operating Services.
3. Define Context Switching.
4. List the conditions that must hold for Deadlock Prevention.
5. Compare and contrast External Fragmentation and Internal Fragmentation.
6. State the various Page Replacement Algorithms used for Page Replacement.
7. Define Seek Time and Latency Time.
8. List common schemes for defining the Logical Structure of a Directory.
9. What is the role of a hypervisor in virtualization?
10. Draw the layers of IOS.

PART - B (5×13 = 65 marks)

11. a. (i) Explain the importance of memory hierarchy. (5)
(ii) Explain Direct Memory Access in detail. (8)
(OR)
11. b. (i) Explain the concept of Multiprogramming and its advantages in brief. (5)
(ii) Explain in detail about the categories of system calls. (8)
12. a. (i) Discuss in brief about process states and PCB. (5)

- (ii) Write about semaphores, their usage and implementation to avoid busy waiting.

(8)

(OR)

12. b. Explain the Banker's algorithm for deadlock avoidance.

Assume that there are 5 processes, P0 through P4, and 4 types of resources. At T0 we have the following system state: Max Instances of Resource Type A = 3 (2 allocated + 1 Available) Max Instances of Resource Type B = 17 (12 allocated + 5 Available) Max Instances of Resource Type C = 16 (14 allocated + 2 Available) Max Instances of Resource Type D = 12 (12 allocated + 0 Available)

1. Create the need matrix (max-allocation)

2. Use the safety algorithm to test if the system is in a safe state or not?

3. If the system is in a safe state, can the following requests be granted, explain why or why not?

a. P1 requests (2,1,1,0)

b. P1 requests (0,2,1,0)

(13)

13. a. Describe the basic concepts about paging and explain the types of Page Table Structure.

(13)

(OR)

13. b. Explain the concept of demand paging and how demand paging is implemented with virtual memory.

(13)

14. a. Suppose that the disk drive has 5000 cylinders number 0 to 4999. The drive is currently serving a request at cylinder 143 and the previous request was at 125, the queue of the pending request in FIFO order is: 86,1470,913,1174,948,1509,1022,1750,130 starting from the current head position, what is the total distance (cylinders) that the disk arm moves to satisfy all the pending requests for each of the disk scheduling algorithms.

(i) SSTF

(ii) SCAN

(iii) LOOK

(iv) C-LOOK.

(13)

(OR)

14. b. (i) Explain in detail about free space management with neat diagram.

(6)

(ii) Describe the two level and tree type directory structures in detail.

(7)

15. a. Explain how virtual machines work with the help of a diagram. Discuss the interaction between host OS, hypervisor, and guest OS.

(13)

(OR)

15. b. Explain in detail about IOS architecture and SDK framework.

(13)

PART - C ($1 \times 15 = 15$ marks)

16. a. Consider the following set of processes, with the length of the CPU burst time given in millisecond

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

- A) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).
- B) What is the turnaround time of each process for each of the scheduling algorithms in part A?
- C) What is the waiting time of each process for each of these scheduling algorithms?
- D) Which of the algorithms results in the minimum average waiting time (over all processes)?

(15)

(OR)

16. b. Consider the following page reference string:

1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

How many page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, or seven frames?

Remember all frames are initially empty, so your first unique pages will all cost one fault each.

- (i) LRU replacement
- (ii) FIFO replacement
- (iii) Optimal replacement.

(15)

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Question Paper Code: 10991

B.E/B.Tech. DEGREE EXAMINATIONS, APRIL / MAY 2025

Fourth Semester

Computer Science and Engineering

(Common to Information Technology)

22CS495 – Java Programming

(Regulation 2022)

Time: Three hours

Maximum:100 marks

Answer ALL questions.

PART - A (10×2 = 20 marks)

1. Write a Java program that demonstrates the use of different primitive data types (int, float, char, boolean). Also, print the size of each type using sizeof() equivalent in Java.
2. Write a Java program to demonstrate the use of switch statement to find the day of the week based on an integer input (1 for Monday, 2 for Tuesday, etc.).
3. How does the finalize() method work in Java? What role does it play in object destruction?
4. Write a Java program to create a parameterized constructor that initializes instance variables, and show how to create an object using that constructor.
5. Define runtime polymorphism. How does Java achieve runtime polymorphism through method overriding?
6. Complete the following code to define a package and a class Calculator with an add method.
package _____;
public class Calculator {
//add method
}
7. Write a program to serialize an object of class Student into "student.ser" file.
8. Explain the use of ActionListener interface.
9. What is the difference between a Statement and a PreparedStatement in JDBC?
10. Write a SQL query using GROUP BY and any aggregate function like SUM() or COUNT().

PART - B (5×13 = 65 marks)

11. a. (i) A weather monitoring system records daily temperatures. If a city experiences three consecutive days of increasing temperatures, an alert is

issued. Unary operators should be used to track day-to-day temperature changes.

Use Scanner for temperature input.

Apply unary increment/decrement operators to track trends.

Use logical operators to detect temperature rise alerts. (7)

(ii) Discuss the precedence and associativity of operators in Java. Provide examples demonstrating their usage. (6)

(OR)

11. b. You are developing a student grading and rewards system for an online platform. Implement the following scenarios using the appropriate selection statements:

Grade Assignment: If the student's score is 90 or more, assign grade A; between 80-89, assign grade B; between 70-79, assign grade C; below 70, assign grade D. Use: if-else-if.

Reward Eligibility: If the student's score is above 85, grant a special reward; if the score is 85 or less, do not grant any reward. Use: if-else.

Course Level Eligibility:

The system needs to assign the appropriate course level based on the student's year of study. If the student is in the 1st year, assign them to Beginner Level; in the 2nd year, assign to Intermediate Level; in the 3rd year, assign to Advanced Level. For any other year, print Invalid year. Use: switch. (13)

12. a. (i) You are developing a text editor that processes user input. The application should accept a string and perform the following tasks using String class methods:

1. Find the length of the string.

2. Find the index of a specific character/word.

3. Extract a character at a given index.

4. Replace a specific word and convert the string to uppercase and lowercase.

5. Compare two strings using compareTo() and check if the string contains a word. (9)

(ii) Demonstrate how you can iterate through a 2D array using nested loops. (4)

(OR)

12. b. (i) Discuss the use of **static members** in Java. (3)

(ii) Discuss about the types of constructors in java. You are tasked with creating a Book class in Java. The class should support constructor overloading: One constructor that takes title and author as parameters.

Another constructor that also includes price and ISBN as parameters.

Implement methods to display book details.

Demonstrate the creation of Book objects using both constructors.

Discuss how constructor overloading improves code flexibility. (10)

13. a. Discuss the concept of packages in Java. How do you organize classes and interfaces into packages? Explain how to define a package and import it into other classes. How does package-level access protection work using the private, protected, and public access modifiers? (13)

(OR)

13. b. Explain the concept of exception handling in Java. designing an online shopping system that connects to a database to fetch product details. The system needs to handle potential database connection errors.

- If the database connection fails, throw a `DatabaseConnectionException`.
- If the product ID entered by the user is invalid, throw an `InvalidProductIDException`.
- If the query to fetch product details results in an error, throw a `QueryExecutionException`.

Use try-catch blocks to handle these exceptions and provide appropriate messages.

(13)

14. a. (i) You are tasked with creating a Simple Calculator using the AWT (Abstract Window Toolkit) in Java. The calculator should be able to perform basic operations such as addition, subtraction, multiplication, and division.

- Design the user interface with buttons for numbers (0-9) and operators (+, -, *, /), along with a `TextField` to display results.
- Implement event handling for each button so that the corresponding arithmetic operation is performed when clicked.
- Use Layout Managers to organize the buttons and `TextField` in the GUI.
- Ensure the program uses Event Delegation Model to handle button clicks and display results.

(8)

(ii) Discuss the types of layout managers provided by AWT, including `FlowLayout`, `GridLayout`, `BorderLayout`, and `CardLayout`, with examples of when to use each.

(5)

(OR)

14. b. (i) What is thread synchronization? Why is it necessary in multithreaded programming?

(7)

(ii) Create a Java program that serializes and deserializes an `Employee` object using `ObjectOutputStream` and `ObjectInputStream`.

(6)

15. a. Design a hotel reservation system that tracks guests, rooms, reservations, and payments.

Tables:

Guests: `GuestID`, `Name`, `Email`, `Phone`, `Address`

Rooms: `RoomID`, `RoomType`, `PricePerNight`, `Status`

Reservations: `ReservationID`, `GuestID`, `RoomID`, `CheckInDate`, `CheckOutDate`, `Status`

Payments: `PaymentID`, `ReservationID`, `PaymentDate`, `Amount`, `PaymentMethod`

Tasks:

Create tables. Insert a new guest, room, and reservation.

Retrieve reservations for a guest, with room and payment details.

Update room status when confirmed or cancelled.

Delete reservation and associated payment.

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Create a stored procedure for total payment calculation.
Describe table relationships, foreign keys, cascading deletes, and the use of stored procedures.

(13)

(OR)

15. b. Explain the structure of the Java Collections Framework.

A) What is a Collection in Java?

B) Explain the interfaces: List, Queue, and Set.

C) Write about implementation classes like ArrayList, PriorityQueue, HashSet, or TreeSet.

Illustrate your answer with suitable code snippets and scenarios.

(13)

PART - C (1×15 = 15 marks)

16. a. You are developing an airline booking system that needs to interact with an Oracle database. The system will manage flights, passengers, and bookings. You need to implement various functionalities like adding new passengers, booking flights, and updating the status of a booking.

Database Tables:

- Flights: FlightID, FlightNumber, DepartureDate, ArrivalDate, Origin, Destination
- Passengers: PassengerID, Name, Age, Email
- Bookings: BookingID, PassengerID, FlightID, BookingDate, SeatNumber, Status

Tasks:

- Connecting to the Database: Establish a JDBC connection using Type 4 JDBC driver and Oracle database.
- Retrieve Data: Retrieve a list of available flights based on the origin and destination.
- Update Data: Use PreparedStatement to update the seat number of a booking.
- Delete Data: Use PreparedStatement to delete a booking record based on BookingID.

Stored Procedure: Create a stored procedure that updates the flight status based on the flight number. Use CallableStatement to call the stored procedure from Java.

(15)

(OR)

16. b. (i) You are developing a simple Employee Hierarchy for an organization. The system includes:

A Person class with attributes such as name and address. An Employee class that extends Person and adds an employeeID and salary. A Manager class that extends Employee and adds a department attribute. Demonstrate Multilevel Inheritance by creating an object and explain how Multilevel Inheritance works.

(8)

(ii) Explain method overloading in Java. How is method overloading different from method overriding?

(7)